EXPLAINABLE BEHAVIORAL ANOMALY DETECTION FOR CONTAINERIZED APPLICATIONS

PROCESSPULSE-C

BHUVANESHWARAN S 24BDS0206
GURUPRASAD V 24BCE0761
RAGUNANDHAN T 24BCE0721

PROJECT OVERVIEW

PROJECT MOTIVATION

- Rapid adoption of containerized apps increases monitoring and security complexity
- Traditional methods miss sophisticated behavioral threats in containers
- Critical vulnerabilities emerge across cloud-native infrastructure

REVIEW 1 OBJECTIVES

- Define behavioral anomaly detection in container environments
- Survey container security and explainable AI literature
- Identify the gap in interpretable detection systems
- Combine system-call analysis with explainable ML

TECHNICAL BACKGROUND

01 CONTAINERIZATION & PROCESS MONITORING

- Linux namespaces and cgroups provide lightweight OS-level virtualization
- Shared host kernels create security challenges unlike virtual machines
- System calls, resource use and network activity reveal behavior
- Baseline profiles enable anomaly detection

CONTAINERS

SYSCALLS

02 ML & EXPLAINABLE AI

- ML detects subtle deviations in high-dimensional telemetry
- XAI opens black-box decisions to interpretation
- Analysts can validate alerts and understand threats

ML

XAI

PROBLEM STATEMENT

DETECTION CHALLENGES

- Subtle anomalies evade signature-based detection
- Adaptive detection is needed for zero-day attacks
- Noisy runtime telemetry obscures anomalies
- Dynamic orchestration shifts baselines constantly

CURRENT LIMITATIONS

- Black-box models lack analyst transparency
- False positives remain high in production
- Alerts provide no actionable explanation
- The alert-to-remediation gap slows response

LITERATURE SURVEY

CONTAINER SECURITY & RUNTIME MONITORING

SYSDIG FALCO (2016) Rule-based syscall monitoring
ABED ET AL. (2015) Syscall-frequency intrusion detection
LIN ET AL. (2018) Namespace isolation and seccomp profiles
COMBE ET AL. (2016) Vulnerabilities and attack surfaces

ANOMALY DETECTION & EXPLAINABILITY

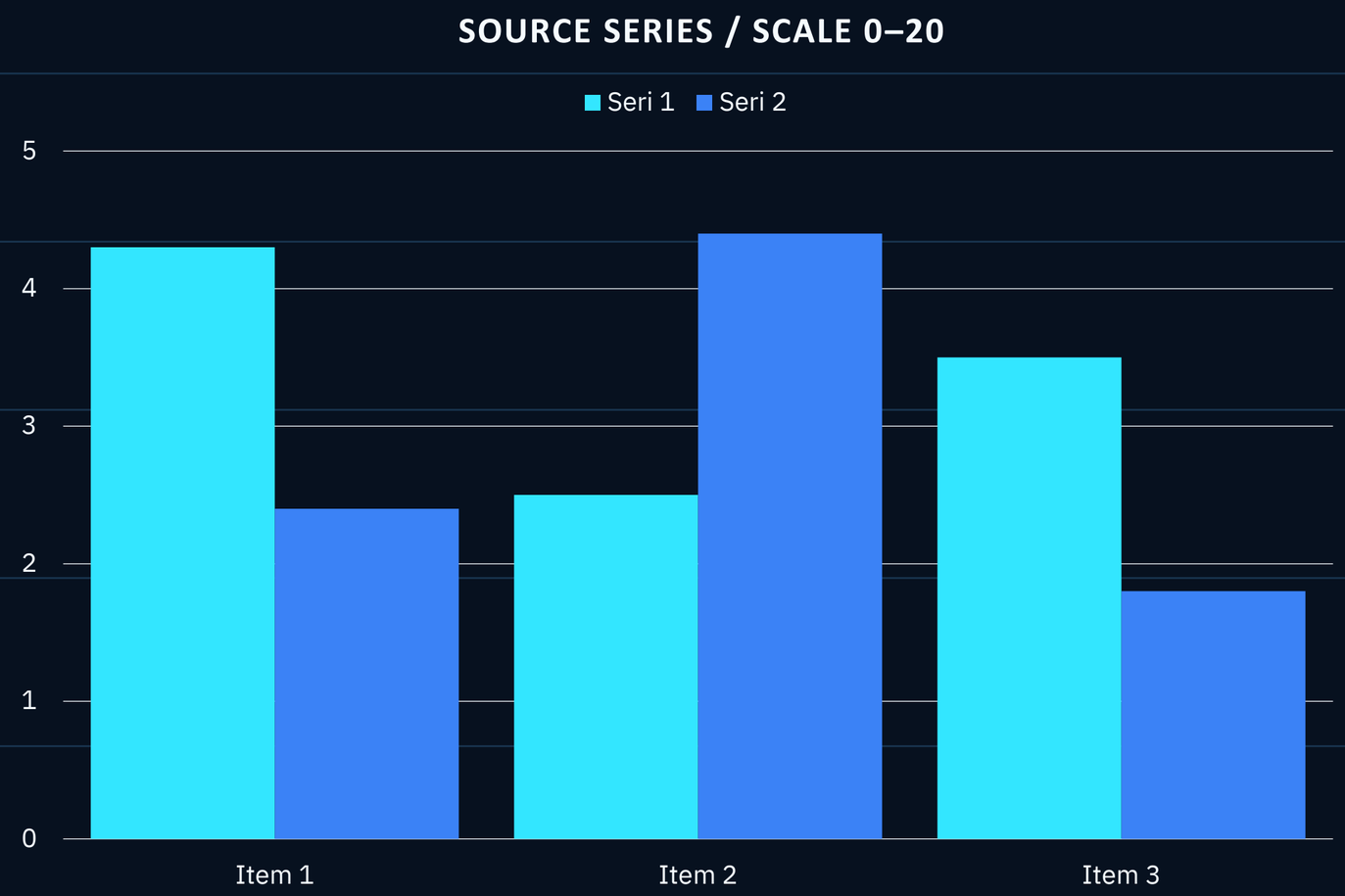
CREECH & HU (2014) Semantic-syscall LSTM host IDS
DU ET AL. (2017) DeepLog for system-log anomalies
RIBEIRO ET AL. (2016) LIME local explanations
LUNDBERG & LEE (2017) SHAP unified feature attribution

LITERATURE ANALYSIS

QUALITATIVE MAPPING

Publications are categorized across Container Monitoring, Anomaly Detection, Explainable AI, and Integrated Solutions.

This chart preserves the source series and is a qualitative assessment — not experimental data.



RESEARCH GAP & OBJECTIVES

THE GAP

- No integrated behavioral telemetry, real-time detection and explainability
- Tools address detection or explanation — rarely both in container context
- Analysts lack actionable incident-response insight

PROJECT OBJECTIVES

01 Design the ProcessPulse-C monitoring framework
02 Derive features from system calls and resource use
03 Develop ML-based anomaly detection for containers
04 Generate explainable alerts for response and root-cause analysis

PROPOSED METHODOLOGY

PHASE 01 / COLLECT & PREPARE

TELEMETRY COLLECTION
System calls, resource metrics, network activity via eBPF

PREPROCESSING PIPELINE
Cleaning, normalization and noise filtering

FEATURE ENGINEERING
Sequences, frequency patterns and temporal features

PHASE 02 / DETECT & EXPLAIN

BASELINE CONSTRUCTION
ML profiles of normal behavior

ANOMALY DETECTION
Real-time deviation detection

EXPLANATION GENERATION
Human-readable SHAP/LIME evidence

ALERT OUTPUT
Context for analyst review

SYSTEM ARCHITECTURE

DATA COLLECTION & PROCESSING

CONTAINERIZED APPLICATION LAYER
Docker / Kubernetes runtime

RUNTIME TELEMETRY COLLECTORS
eBPF syscalls / resource metrics

FEATURE PROCESSING MODULE
Profiles / temporal encoding

DETECTION & EXPLAINABILITY

ANOMALY DETECTION ENGINE
ML classifier / real-time inference

EXPLAINABILITY MODULE
SHAP / LIME attribution

DASHBOARD & ALERT INTERFACE
Actionable anomaly reporting

PROPOSED EVALUATION

EVALUATION STRATEGY

Representative container workloads with injected cryptomining, privilege escalation and data-exfiltration anomalies.

PROPOSED METRICS

DETECTION
Precision · Recall · F1-Score

OPERATIONS
False-positive rate · Detection latency

EXPLAINABILITY
Explanation usefulness score

EXPECTED OUTCOMES

- High accuracy with minimal false positives
- Production-suitable real-time latency
- Expert-validated, interpretable explanations

EXPECTED CONTRIBUTIONS & NEXT STEPS

EXPECTED CONTRIBUTIONS

- Explainable behavioral anomaly workflow using eBPF and interpretable ML
- Modular real-time security architecture with human-readable evidence
- Bridge detection accuracy and operational transparency

PROJECT PLAN

01 IMPLEMENTATION eBPF probes, features, ML integration
02 DATASET benign workloads and simulated attacks
03 EXPERIMENTS accuracy, explanations, overhead
04 REFINEMENT iterate from empirical findings

REFERENCES

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THANK YOU

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OPERATING SYSTEMS / PROJECT REVIEW